p9-a review changelog (2026-10-22)

Lessons: `react-mastery/09-advanced-rendering-and-ecosystem/01-spa-ssr-ssg-streaming.md`, `02-hydration-internals-deep-dive.md`. Scratch (all runnable): `tools/sandbox/scratch/p9-a/` (s1-s4, ppr, b1, h1-h6, ser, toy.mts, c3.test.tsx). Runtime: react/react-dom 19.3.x, Node 24, jsdom, Vitest 3. Note: react.dev returns HTTP 200 for non-existent pages, so every react.dev URL was checked by page content, not status code.

Lesson 01 - SPA, SSR, SSG, streaming

#	Claim	How checked	Result	Change
1	Streamed HTML shape: pending boundary, swap script	s1.mjs (renderToPipeableStream, 2 slow boundaries, chunk timestamps)	Confirmed, and shape is richer than lesson implied: `<!--$?-->` `<template id="B:0">`, `<div hidden id="S:0">`, `$RC("B:0","S:0")`, `$RB/$RV` batching, `<link rel=preload>` in head, `<script async id=_R_>`	Replaced the "never look at markup" warning with the observed 4-chunk output (labelled implementation detail)
2	19.2 batching of Suspense reveals, ~2.5 s cutoff	19.2 blog text + `$RC` source in the output (`2300>a&&2E3<a`)	Confirmed	Added to 2.3 and 3.2
3	`onShellReady` / `onAllReady` / `onShellError` / abort behaviour	s2.mjs, s3.mjs; react.dev renderToPipeableStream	Confirmed: shell throw -> onError then onShellError, no bytes; suspending shell + abort -> onShellError; boundary + abort -> onAllReady still fires, `$RX` marks boundary client-render	Corrected 3.1 #2 (a suspending shell without boundary waits, it is not an immediate onShellError)
4	Status code after boundary error ("200", B1, 5.1 predict)	s2/s3	**Corrected/nuanced**: a synchronous throw inside a boundary calls onError BEFORE onShellReady, so the lesson's own `didError ? 500 : 200` handler sends 500; a later rejection arrives after pipe -> 200	Edited 5.1 note (1), predict, and answer B1
5	`renderToString` with Suspense	s1/s2	Confirmed it cannot wait; pending boundary -> `<!--$!-->` + template with explicit message + fallback; throw outside boundary throws	Added to misconceptions table row
6	`prerender` returns `{ prelude }`	react.dev prerender page, s1/s2/ppr	**Corrected**: returns `{ prelude, postponed }`; prelude is Web Stream; Node variant `prerenderToNodeStream`; aborted prerender -> fallback in prelude + non-null postponed; `nonce` option not available	Edited 3.1 #1/#5, 3.4, 5.2 note
7	19.2 resume APIs / PPR (marked volatile)	react.dev prerender page, 19.2 blog, ppr.mjs (prerender+abort then `resume`)	Confirmed and executed: resumed stream contains only the remaining boundary (`$RC("B:0","S:0")`), prelude already ends `</body></html>`; names/locations match (resume, resumeToPipeableStream in server; resumeAndPrerender(+ToNodeStream) in static)	Rewrote 3.4 paragraph and 3.6 with the verified mechanism, removed "volatile" hedge; Next.js: PPR is default with Cache Components (nextjs.org docs)

#	Claim	How checked	Result	Change
8	Web Streams APIs in Node	19.2 blog; s1 ran `renderToReadableStream` in Node	Confirmed; Node Streams still recommended	Added to 3.1 #1 and links table
9	nginx `X-Accel-Buffering: no`	nginx proxy module docs	Confirmed	Removed warning, quoted
10	Compression middleware `res.flush()`	expressjs/compression README	Confirmed for Express `compression` (other libs unverified, hedged)	Edited 3.7
11	Dynamic rendering / bot-only onAllReady "cloaking"	Google Search Central dynamic-rendering page (partial text only)	Hedged: Google calls dynamic rendering a workaround, not cloaking when content similar. Crawler JS-rendering claim in 2.7 not independently verified	Reworded 2.7 row and 5.1 note (3)
12	Islands (Astro) / resumability (Qwik)	Qwik docs ("Qwik applications don't hydrate; they resume"); Astro islands page loaded but text extraction poor	Qwik confirmed; Astro only lightly (page exists, concept standard)	Replaced the warning with source attribution
13	`serializeForScript`	ser.mjs	**Corrected**: lesson source contained raw U+2028/2029 characters inside regex literals (invisible, and rendered as spaces); function otherwise works (no `</script>`, round-trips). Fixed to ` / ` escapes; C2 parenthetical "JSON escapes `<` back to `<`" was garbled	Edited 5.7 and C2
14	Paywall / AsyncLocalStorage ⚠	Google paywalled-content structured data page; Node docs	Hedged	Replaced "flexible sampling ⚠" with structured-data reference; ALS support on edge runtimes hedged
15	Link list "unverified" items	content check of all react.dev/web.dev/MDN/GitHub URLs; repo paths exist	Confirmed	Removed unverified markers, added resume/resumeAndPrerender links

Unresolved (01): (a) the 4.1 timing table is illustrative arithmetic, not measured; (b) "major crawlers execute JavaScript with delays" not independently re-verified (hedged to "check current Google docs"); (c) CDN/serverless buffering behaviour per vendor (hedged, not tested); (d) Content-Length forces buffering (D1 answer) not executed.

Lesson 02 - Hydration internals

#	Claim	How checked	Result	Change
1	Text/structure mismatch = recoverable error, tree regenerated	h1.mjs (jsdom, dev and NODE_ENV=production)	Confirmed. Dev message "Hydration failed because the server rendered text/HTML didn't match the client. As a result this tree will be regenerated on the client..." with `+ / −` diff; prod "Minified React error #418" (args text/HTML); errorInfo keys = ["componentStack"]	Added real messages to 2.2, 3.2, 3.7, 5.1, Symptom A

#	Claim	How checked	Result	Change
2	Attribute mismatch semantics	h1.mjs	**Corrected/strengthened**: attribute mismatch is dev `console.error` ("won't be patched up") with a diff, DOM keeps server value, **not** delivered to onRecoverableError, silent in production	Edited 2.2, 3.1 #3, 3.7 taxonomy + notes, 5.1 notes (monitoring caveat), C3 note
3	Blast radius = nearest Suspense boundary above, else root	h2.mjs (node identity checks)	Confirmed: with boundary `<h1>` / `<footer>` nodes kept; without, all replaced	Added executed evidence
4	`suppressHydrationWarning`: one level deep, no patching	h1.mjs dev+prod; react.dev hydrateRoot	Confirmed: text and attribute suppressed, server value kept in DOM; suppress on parent does NOT cover mismatching child (full error)	Edited 2.5, 5.6 (replaced warning marker), A5 consistent
5	Text separators `<!-- -->` and marker comments	s1/h1, 19.3 output	Confirmed `Hello <!-- -->Ada`; markers `$ $? $! /$` confirmed; extra `<!--$~-->` (queued for batched reveal) exists in 19.2+	Edited 3.4, removed warning
6	Event capture and replay (3.6, 4.1, Q6)	h3/h4 + reading react-dom-client.development.js `dispatchEvent`, `discreteReplayableEvents`, `queueIfContinuousEvent`	**Corrected**: discrete events on a dehydrated boundary cause synchronous hydrate-then-dispatch (executed: handler ran inside the same dispatch, boundary B hydrated before A, C); if boundary can't hydrate (lazy chunk pending) the click was dropped; focusin/dragenter/mouseover/pointerover are queued and replayed (so "continuous events not replayed" was too broad; pointermove is not)	Rewrote 3.6, 4.1 steps 4-5, gantt label, Q6, checkpoint line
7	useSyncExternalStore hydration (B3, 5.3)	h2.mjs	Confirmed: renders server snapshot (1024) then client snapshot (390), no error	none
8	useId prefix / format	executed (`_R_0_`), 19.2 blog	Updated with 19.0 `:r:`, 19.1 `«r»`, 19.2 `_r_` history	Edited 3.8
9	useLayoutEffect warns on server ⚠	s4.mjs	No warning on 19.3	Edited 3.8, removed marker
10	Portals ⚠	s4.mjs	`renderToString` throws "Portals are not currently supported by the server renderer..."	Edited 3.8
11	React 19 skips unexpected head/body tags	h5.mjs (`hydrateRoot(document)` with extra style in head, extra divs first/last in body); 19 blog	Confirmed, no errors	Added executed evidence
12	Strict Mode + hydration	h5.mjs; 19.3 changelog (#35961)	Confirmed 19.3 double-invokes effects during hydration (`setup, cleanup, setup`)	Edited 3.9; softened "twice as visibly"
13	Toy hydrator and predictions (3.3, B1, 6.1)	toy.mts via node --experimental-strip-types; tsc --strict	Confirmed: "expected ", "extra server nodes inside ", "expected "; only tsc error is harness `jsdom` types	Added "executed" note

#	Claim	How checked	Result	Change
14	onRecoverableError default	read react-dom-client source (defaultOnRecoverableError -> reportGlobalError); react.dev	Confirmed; also fires for server-error boundaries (h6.mjs: "Switched to client rendering because the server rendering errored: ...", client retry succeeds)	Edited 5.1 notes
15	C3 Vitest hydration test	c3.test.tsx run under Vitest 3 + jsdom	Works with `IS_REACT_ACT_ENVIRONMENT=true`, `act` from react; detects text mismatch; lesson sample lacked imports	Added imports + note
16	19.3 `browser()`	r193.txt, b1.mjs (use(browser()) -> `<!--$!-->` + fallback, onBrowserBailout(error, {componentStack}))	Confirmed	Added pointer in links section
17	Links	content checks	Confirmed; discussion #37 title matches	Removed unverified markers

Unresolved (02): (a) exact set of events replayed is read from 19.3 source, not documented, and may change (hedged in text); (b) real-browser behaviours (Safari-specific, extension-induced mismatches, whitespace/comment skipping rules in C2) not testable in jsdom - hedged; (c) D1 "wrong product" mechanism remains a plausible-scenario answer (clarified that handlers come from the client's item at that DOM position while attributes stay server-side); (d) "root.render before hydration finishes clears server HTML" (hydrateRoot docs) not added or executed.

Counts

Confirmed ~21 claim groups, corrected ~10 (status code timing, prerender return shape, attribute-mismatch reporting, event replay semantics, shell-suspends behaviour, suppressHydrationWarning child case, useLayoutEffect/portal notes, serializeForScript source, continuous events, useId format), unresolved/hedged 8 (listed above).